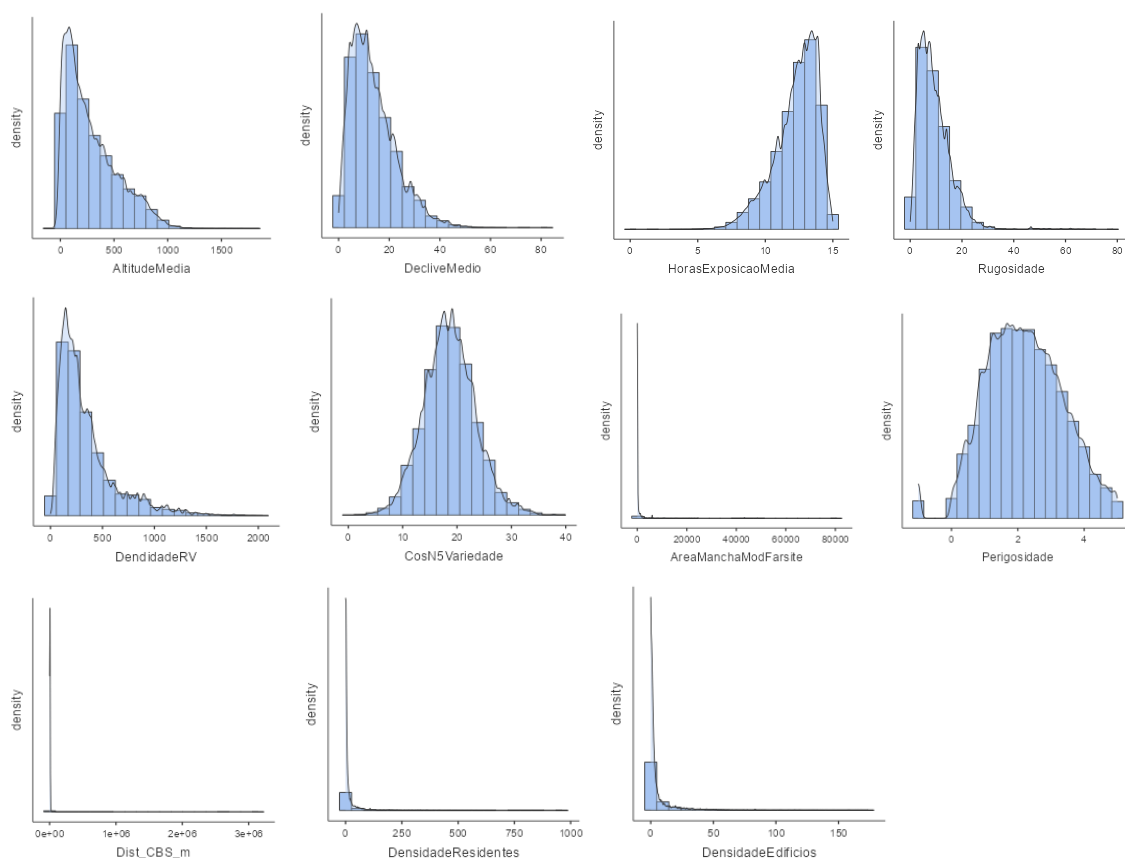


Appendix III – New Data Fields and Descriptive Statistical Overview

Based on the initial classification of new fields we present a descriptive analysis of the first two groups social and physical data.

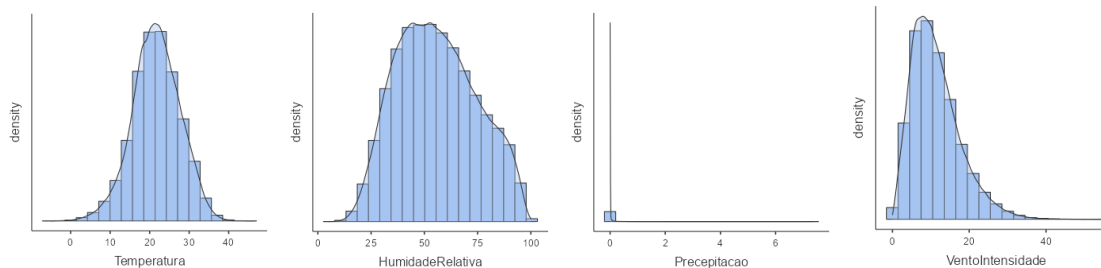
	N	Missing	Mean	SE	Median	SD	Variance	Minimum	Maximum	Skewness		Kurtosis	
										Skewness	SE	Kurtosis	SE
AltitudeMedia	809826	165	284.61	0.26771	217.24	240.92	58040.82	-64.21	1848.80	0.9715	0.00272	0.330	0.00544
DecliveMedio	809826	165	13.69	0.00979	11.83	8.81	77.62	0.00	82.34	1.0091	0.00272	0.996	0.00544
HorasExposicaoMedia	809967	24	12.28	0.00176	12.59	1.59	2.52	0.00	15.00	-0.9359	0.00272	0.724	0.00544
Rugosidade	809967	24	9.30	0.00686	8.05	6.18	38.15	0.00	78.28	1.9285	0.00272	9.637	0.00544
DensidadeRV	809849	142	338.92	0.30194	255.87	271.72	73833.72	0.00	2041.13	1.7721	0.00272	3.601	0.00544
CosN5Variedade	809858	133	18.62	0.00526	18.55	4.73	22.39	1.00	39.88	0.1546	0.00272	0.275	0.00544
AreaManchaModFarsite	198809	611182	637.98	8.29518	28.79	3699.10	1.37e+7	1.00	80409.60	13.5592	0.00549	229.482	0.01099
Perigosidade	809991	0	2.21	0.00127	2.16	1.15	1.31	-1.00	5.00	0.0877	0.00272	-0.303	0.00544
Dist_CBS_m	809991	0	4374.19	6.40907	3805	5768.13	3.33e+7	0	3141387	316.3321	0.00272	157746.284	0.00544
DensidadeResidentes	796826	13165	16.13	0.04518	3.62	40.33	1626.71	0.00	961.31	5.6504	0.00274	42.730	0.00549
DensidadeEdificios	799338	10653	5.47	0.01292	1.63	11.56	133.52	0.00	173.41	4.9115	0.00274	35.259	0.00548

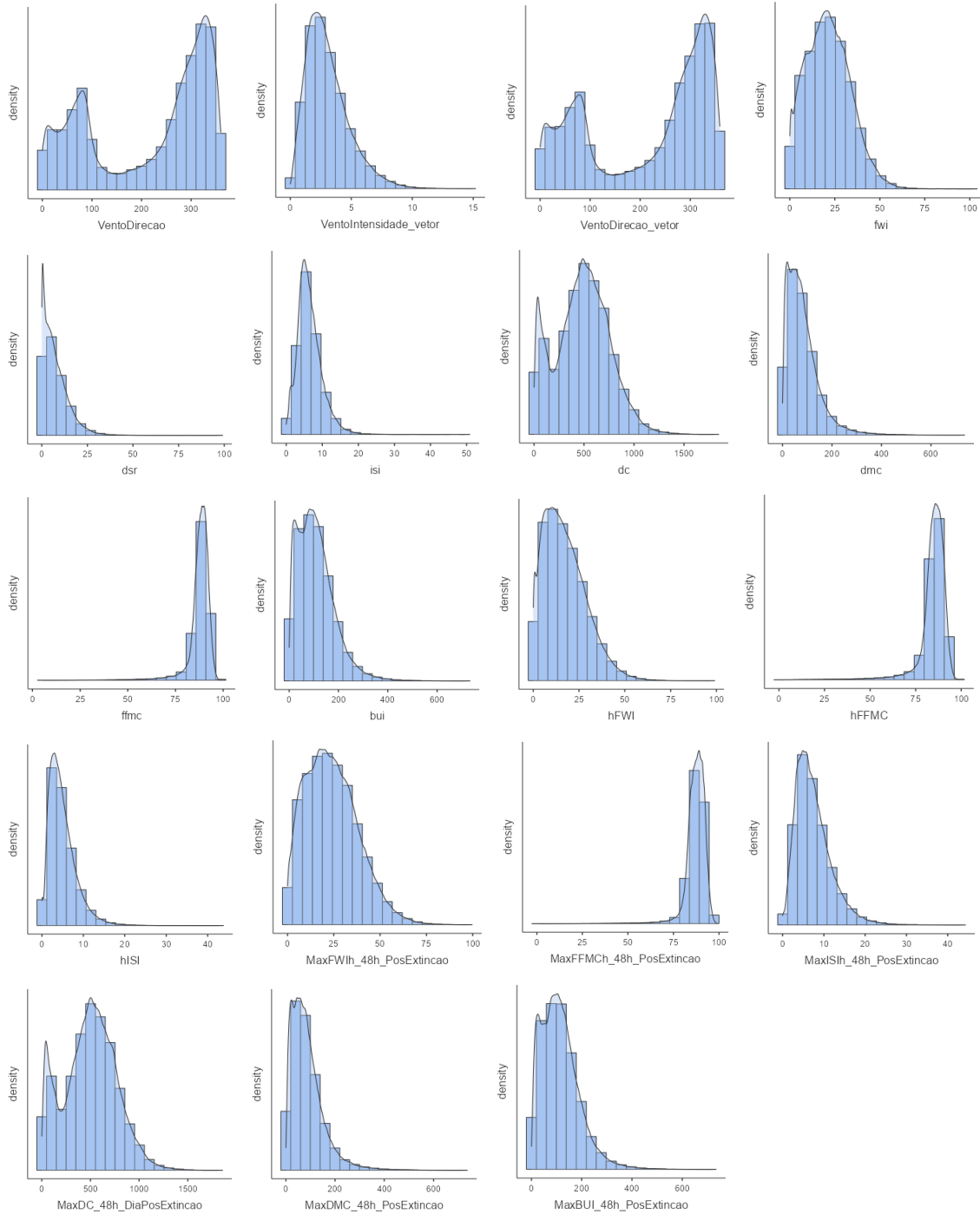


The data shows that for altitude (AltitudeMedia), with a mean of 284.61m and a standard deviation of 240.92, the slight positive skewness and low kurtosis suggest the median is more appropriate to use due to a tail towards higher values. Similarly, for slope (DecliveMedio), the mean slope is 13.69% with a standard deviation of 8.81, but the positive skewness and near-normal kurtosis indicate the median is a better representation. Exposure hours (HorasExposicaoMedia) have a mean of 12.28 hours and a standard deviation of 1.59, with negative skewness and slight kurtosis, suggesting the median is better due to the tail towards

lower exposure hours. Terrain rugosity (Rugosidade), with a mean of 9.30 and a standard deviation of 6.18, shows high positive skewness and kurtosis, making the median more suitable due to the presence of outliers. For road density (DensidadeRV), with a mean of 338.92 and a standard deviation of 271.72, the high skewness and moderate kurtosis also favor using the median. In contrast, variability on land use (CosN5Variedade), with a mean of 18.62 and low skewness and kurtosis, is better represented by the mean. However, the size of fuel model area (AreaManchaModFarsite), which has a mean of 637.98 units and a very high standard deviation of 3699.10, along with extreme skewness and kurtosis, is better represented by the median due to significant outliers. The danger index (Perigosidade) has a mean of 2.21 and low skewness and kurtosis, making the mean suitable. Distance to fire brigades (Dist_CBS_m), with a mean of 4374.19 meters and a very high standard deviation of 5768.13, shows extreme skewness and kurtosis, indicating that the median is more appropriate. Lastly, for social information both density of residents (DensidadeResidentes) (mean of 16.13) and building density (DensidadeEdificios) (mean of 5.47) have high skewness and kurtosis, suggesting that the median is better due to the presence of higher values and outliers.

	N	Missing	Mean	SE	Median	SD	Variance	Minimum	Maximum	Skewness		Kurtosis	
										Skewness	SE	Kurtosis	SE
Temperatura	806296	3695	21.6547	0.00671	21.65	6.028	36.3419	-4.93409	46.37	-0.140	0.00273	0.0949	0.00546
HumidadeRelativa	806296	3695	55.8400	0.02074	54.62	18.621	346.7281	4.83346	100.01	0.186	0.00273	-0.7654	0.00546
Precepitacao	806296	3695	0.0142	1.34e-4	0.00	0.120	0.0144	-8.67e-16	7.35	19.603	0.00273	572.0749	0.00546
VentoIntensidade	806296	3695	11.3318	0.00703	10.27	6.312	39.8448	0.00414	54.08	0.975	0.00273	1.2390	0.00546
VentoDirecao	806287	3704	215.5281	0.13238	268.95	118.866	14129.1731	2.06e-4	360.00	-0.489	0.00273	-1.3810	0.00546
VentoIntensidade_vetor	806287	3704	3.0590	0.00194	2.74	1.746	3.0489	0.00621	14.79	1.031	0.00273	1.3512	0.00546
VentoDirecao_vetor	806287	3704	215.6805	0.13349	270	119.865	14367.5488	0	359	-0.505	0.00273	-1.3788	0.00546
fwi	806287	3704	21.5435	0.01340	20.90	12.031	144.7464	3.15e-8	101.26	0.417	0.00273	-0.0205	0.00546
dsr	806287	3704	7.5585	0.00755	5.90	6.779	45.9603	1.44e-15	96.42	1.590	0.00273	4.7763	0.00546
isi	806287	3704	6.4578	0.00376	5.95	3.373	11.3781	1.55e-7	49.51	1.156	0.00273	3.6434	0.00546
dc	806287	3704	485.0279	0.29433	492.32	264.292	69850.3065	0.32922	1798.09	0.138	0.00273	-0.2948	0.00546
dmc	806287	3704	82.1325	0.07212	68.62	64.762	4194.1551	2.77e-5	716.23	1.854	0.00273	5.9668	0.00546
ffmc	806287	3704	87.5541	0.00694	88.62	6.236	38.8838	5.32149	98.92	-3.890	0.00273	24.2077	0.00546
bui	806287	3704	109.0479	0.08247	98.80	74.050	5483.4111	0.00000	713.88	1.155	0.00273	2.4007	0.00546
hFWI	806287	3704	17.3415	0.01272	15.63	11.418	130.3623	6.38e-10	96.37	0.797	0.00273	0.5513	0.00546
hFFMC	806287	3704	84.4498	0.00889	85.76	7.985	63.7600	0.01848	98.93	-3.759	0.00273	22.1789	0.00546
hISI	806287	3704	4.9238	0.00358	4.25	3.216	10.3440	2.86e-9	42.63	1.443	0.00273	3.6495	0.00546
MaxFWIh_48h_PosExtincao	805719	4272	23.9518	0.01553	22.66	13.942	194.3889	4.64e-9	97.00	0.554	0.00273	0.0284	0.00546
MaxFFMCh_48h_PosExtincao	805719	4272	87.2355	0.00638	87.90	5.725	32.7780	1.89429	99.18	-3.749	0.00273	29.3861	0.00546
MaxSIh_48h_PosExtincao	805719	4272	7.4215	0.00476	6.62	4.272	18.2522	2.13e-8	43.11	1.211	0.00273	2.4020	0.00546
MaxDC_48h_DiaPosExtincao	805719	4272	497.5105	0.29603	506.64	265.723	70608.7166	0.32922	1807.73	0.110	0.00273	-0.2969	0.00546
MaxDMC_48h_PosExtincao	805719	4272	86.8572	0.07331	73.67	65.808	4330.7507	0.04299	717.36	1.802	0.00273	5.6903	0.00546
MaxBUI_48h_PosExtincao	805719	4272	114.5755	0.08341	105.04	74.872	5605.8229	0.06482	714.88	1.112	0.00273	2.2835	0.00546



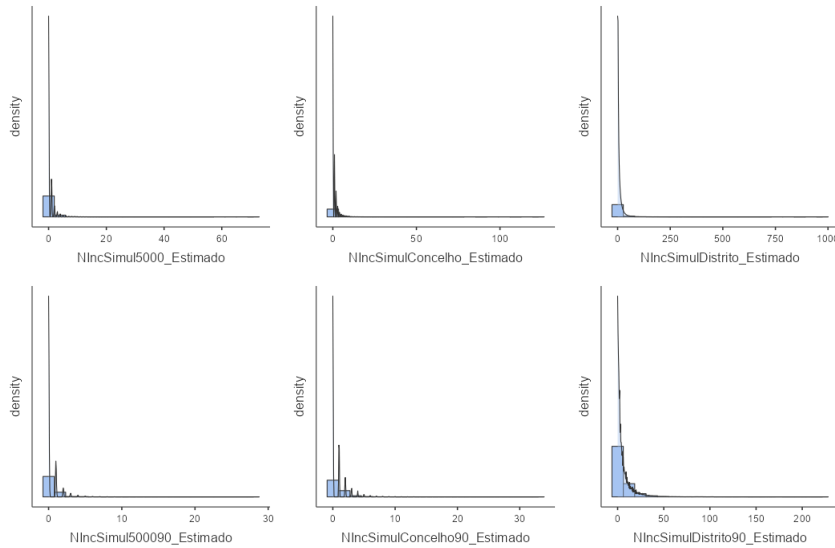


The data on meteorological group reveals that for temperature (Temperatura), with a mean of 21.65°C and a standard deviation of 6.03, the slight negative skewness and low kurtosis suggest the median is appropriate due to the slight tail towards lower temperatures. For relative humidity (HumidadeRelativa), the mean is 55.84% with a standard deviation of 18.62. The positive skewness and negative kurtosis indicate that the mean is a reliable measure, though the median could also be considered due to the tail towards higher humidity values. Precipitation (Precipitação) has a mean of 0.0142 mm and a very low

standard deviation of 0.12, but with extreme positive skewness and kurtosis, indicating the presence of outliers. Hence, the median is a more suitable representation. For wind intensity (VentoIntensidade), the mean wind speed is 11.33 km/h with a standard deviation of 6.31, and moderate positive skewness along with slight leptokurtosis suggests the median is more appropriate due to a tail towards higher wind speeds. Wind direction (VentoDirecao) and wind direction resultant for all time of fire (VentoDirecao_vetor) are not adequate to be analysed as linear continuous variable despite showing a slight negative skewness and negative kurtosis, indicating the data is slightly left-skewed and platykurtic. Fire Weather Index (FWI), with a mean of 21.54 and a standard deviation of 12.03, shows slight positive skewness and near-normal kurtosis, indicating the median might be slightly better due to a rightward tail. DSR (Daily Severity Rating) has a mean of 7.56 and a standard deviation of 6.78, with significant positive skewness and high kurtosis, suggesting the median is more appropriate due to extreme values. For Initial Spread Index (ISI), the mean is 6.46 with a standard deviation of 3.37, and both high positive skewness and kurtosis indicate that the median is better due to the presence of outliers. Drought Code (DC) and Duff Moisture Code (DMC) have means of 485.03 and 82.13, respectively, with standard deviations of 264.29 and 64.76. Both show positive skewness and varying kurtosis, suggesting the median is better for representing central tendencies, particularly for DMC with its extreme skewness and kurtosis. For Fine Fuel Moisture Code (FFMC), the mean is 87.55 with a standard deviation of 6.24, but very high negative skewness and extreme kurtosis indicate that the median is more suitable due to significant lower-end outliers. Build-Up Index (BUI), with a mean of 109.05 and a standard deviation of 74.05, has high positive skewness and moderate kurtosis, suggesting the median is better due to a tail towards higher values. Hourly FWI (hFWI) and hourly ISI (hISI) also show high skewness and kurtosis, making the median more appropriate for these variables. Conversely, maximum of hourly FWI in the 48 hours pos fire (MaxFWIh_48h_PosExtincao) and maximum of hourly FWI in the 48 hours pos fire (MaxISih_48h_PosExtincao) have skewness and kurtosis values that indicate a median would better represent the data due to extreme outliers.

Overall, for most variables with high skewness and kurtosis, the median is a more reliable measure of central tendency, especially where significant outliers or non-normal distributions are present.

	N	Missing	Mean	SE	Median	SD	Variance	Minimum	Maximum	Skewness		Kurtosis	
										Skewness	SE	Kurtosis	SE
NIncSimul5000_Estimado	809991	0	0.477	0.00165	0	1.488	2.213	0	71	9.01	0.00272	141.5	0.00544
NIncSimulConcelho_Estimado	803721	6270	0.921	0.00284	0	2.548	6.493	0	123	8.77	0.00273	132.4	0.00546
NIncSimulDistrito_Estimado	803721	6270	8.693	0.02485	3	22.278	496.311	0	974	9.14	0.00273	127.8	0.00546
NIncSimul500090_Estimado	809991	0	0.351	0.00108	0	0.972	0.945	0	28	6.35	0.00272	71.8	0.00544
NIncSimulConcelho90_Estimado	805110	4881	0.670	0.00177	0.00	1.589	2.525	0	33	6.22	0.00273	65.1	0.00546
NIncSimulDistrito90_Estimado	805110	4881	6.227	0.01400	2.00	12.560	157.764	0	222	6.95	0.00273	72.9	0.00546



The data resulting from reanalysis for number of simultaneous fires in radius of 5000 meters (NIncSimul5000_Estimado) shows a mean of 0.477 incidents with a standard deviation of 1.488. The high positive skewness (9.01) and extreme kurtosis (141.5) indicate a distribution with many low values and a few extreme outliers, making the median of 0 a more appropriate measure of central tendency. For number of simultaneous fires in same municipality (NIncSimulConcelho_Estimado), the mean is 0.921 incidents, with a standard deviation of 2.548. The skewness (8.77) and kurtosis (132.4) are both very high, suggesting that the data is heavily skewed to the right with a long tail and extreme outliers. Thus, the median of 0 incidents is a better representation. The number of simultaneous fires in same district (NIncSimulDistrito_Estimado) variable has a mean of 8.693 incidents and a standard deviation of 22.278. The skewness (9.14) and kurtosis (127.8) are very high, indicating that the data has a long right tail with significant outliers. Given this, the median of 3 incidents provides a better measure of central tendency.

For number of simultaneous fires in radius of 5000 meters in the first 90 minutes (NIncSimul500090_Estimado), the mean is 0.351 incidents, with a standard deviation of 0.972. The positive skewness (6.35) and high kurtosis (71.8) suggest that the distribution is heavily right skewed with outliers, making the median of 0 incidents more appropriate.

Number of simultaneous fires in same municipality in the first 90 minutes (NIncSimulConcelho90_Estimado) shows a mean of 0.670 incidents and a standard deviation of 1.589. The data has high skewness (6.22) and kurtosis (65.1), indicating a distribution with a long right tail and several outliers. Therefore, the median of 0.00 incidents is more suitable for representing this data. Finally, number of simultaneous fires in same district in the first 90 minutes (NIncSimulDistrito90_Estimado) has a mean of 6.227 incidents with a standard deviation of 12.560. The skewness (6.95) and kurtosis (72.9) are relatively high, pointing to a right-skewed distribution with some extreme values. The median of 2.00 incidents is a better measure of central tendency for this variable due to the presence of significant outliers.

In summary, across these variables, the high skewness and kurtosis values suggest that the median is generally a more reliable measure of central tendency than the mean, due to the presence of extreme values and right-skewed distributions.